

ALCOHOL DETECTION & MACHINE SAFETY INTERLOCK

8051-Based Embedded Control System

Project Documentation

AT89C51 • MQ-3 • L293D • LM016L LCD • Embedded C

1. Project Overview

This project implements an embedded safety-interlock system that detects alcohol using an MQ-3 sensor and uses an AT89C51 microcontroller to control a motor-driven machine. The controller continuously monitors the sensor input and selects one of two operating states. In the alcohol-free state, the machine is allowed to run and the buzzer remains OFF. When alcohol is detected, the controller stops the machine through an L293D motor driver, activates a buzzer, and displays an alert on an LM016L LCD.

The project demonstrates practical embedded-system integration across sensing, digital I/O, display interfacing, actuator control and safety-oriented decision logic.

Project Objective

To develop a simple embedded control system in which an alcohol-sensing input determines whether a motor-driven machine is permitted to operate, while providing simultaneous visual and audible status indication.

Core Technologies

Microcontroller	Sensor	Actuation	Interface
AT89C51 / 8051	MQ-3	L293D + Motor	LM016L LCD + Buzzer

2. System Architecture

The system follows a sensor–controller–actuator architecture. The MQ-3 sensor supplies the digital input to the AT89C51. The controller evaluates that input and drives the motor driver, buzzer and LCD.

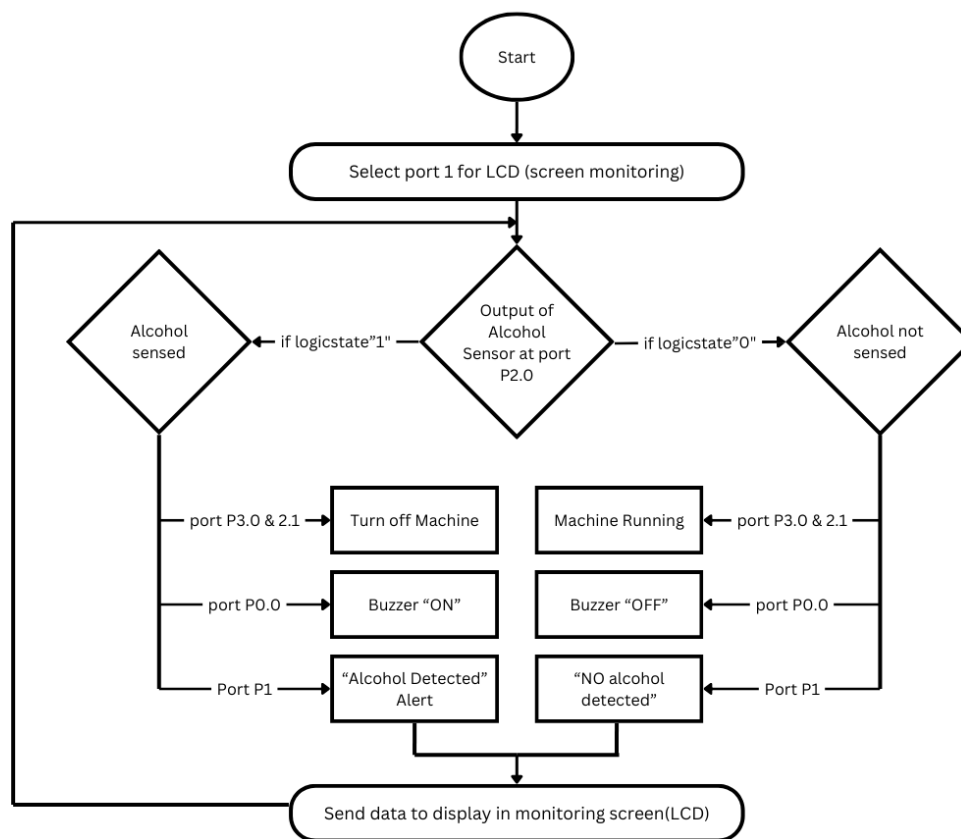


Figure 1. System control flow implemented by the project.

Control Flow

- Initialize the LCD interface.

- Read the MQ-3 sensor state through P2.0.
- Sensor state 0 → run motor, buzzer OFF, display “Alcohol Free”.
- Sensor state 1 → stop motor, buzzer ON, display “Alcohol Detected”.
- Repeat continuously.

3. Hardware Implementation

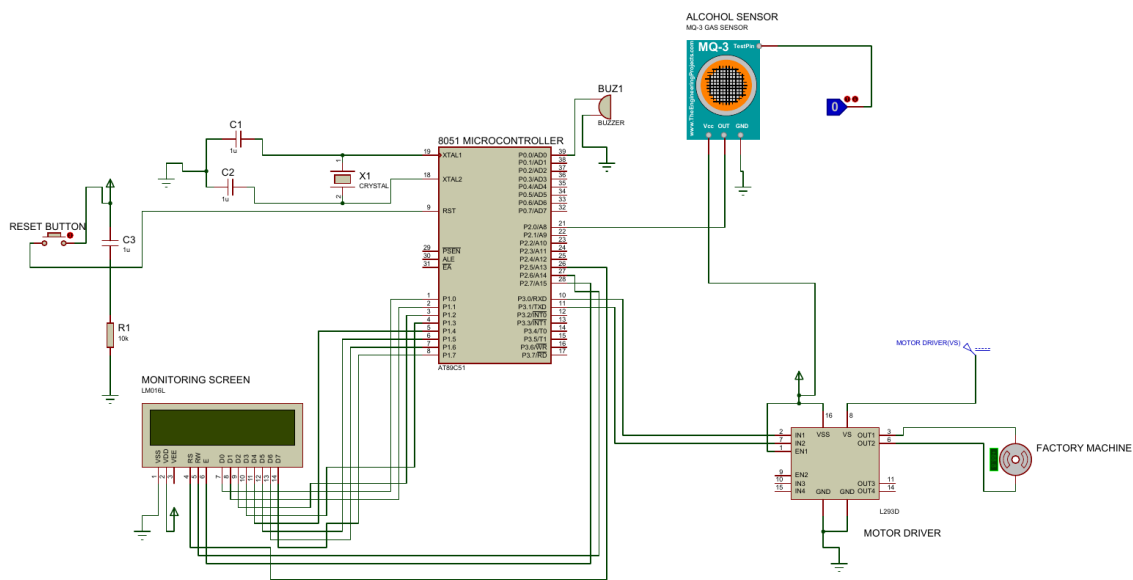


Figure 2. Hardware schematic used for the embedded prototype.

Hardware	Role
AT89C51	Central controller; reads sensor input and generates control signals.
MQ-3 alcohol sensor	Provides the alcohol-detection input.
L293D	Motor-driver interface between the controller and motor.
LM016L LCD	Displays the current system state.
Buzzer	Provides an audible alert during alcohol detection.
Motor / machine load	Actuated according to the detected condition.
Crystal, capacitors & reset network	Support the 8051 clock and reset circuitry shown in the schematic.

4. Microcontroller I/O Mapping

Port / Pin	Code Symbol	Function
P2.0	sensorIN	MQ-3 sensor input
P0.0	buzz	Buzzer output
P3.0	M_a	L293D motor-driver input A
P3.1	M_b	L293D motor-driver input B

Port / Pin	Code Symbol	Function
P1	lcd	LCD 8-bit data bus
P2.5	rs	LCD Register Select
P2.6	rw	LCD Read/Write
P2.7	e	LCD Enable

The supplied program defines these interfaces directly using 8051 bit-addressable pins and a complete Port 1 assignment for the LCD data bus.

5. Software Implementation

The control software is written in C using the 8051-specific reg51.h header. It is organized into hardware-control functions for LCD commands/data, delays, motor control and the main monitoring loop.

Function	Implementation
cmd()	Sends a command to the LCD.
ldata()	Writes a data byte/character to the LCD.
String()	Sends a complete status string to the LCD.
Delay()	Provides software timing delays.
motorRun()	Commands the motor driver with M_a = 1 and M_b = 0.
motorStop()	Sets both motor-driver inputs to 0.
main()	Initializes the LCD and continuously performs sensing and control.

Decision Logic

P2.0	System state	Motor	Buzzer	LCD
0	Alcohol-free state	Motor ON	Buzzer OFF	Alcohol Free
1	Alcohol-detected state	Motor OFF	Buzzer ON	Alcohol Detected

The program explicitly interprets logic 1 as the alcohol-detected state. The actual output polarity of a particular MQ-3 module/comparator arrangement should be verified against the physical module before deployment.

6. Working Principle

Following initialization, the controller enters an infinite monitoring loop. The sensor output is copied into alert. When alert is not equal to 1, motorRun() is executed, the buzzer is cleared, and the LCD displays "Alcohol Free". When alert equals 1, motorStop() is executed, the buzzer is activated, and the LCD displays "Alcohol Detected". The loop then returns to sensor monitoring.

7. Technical Highlights

- 8051 GPIO-based sensor interfacing.
- Digital decision logic for safety-state selection.
- 8-bit LCD interfacing using Port 1 with dedicated control lines.
- L293D-based motor actuation interface.
- Audible and visual status indication.
- Continuous embedded control loop implemented in C.
- Separation of sensing, decision, display and actuator functions.

8. Functional Behaviour

- Alcohol-free input → machine/motor commanded to run + buzzer OFF + LCD displays "Alcohol Free".
- Alcohol-detected input → machine/motor commanded to stop + buzzer ON + LCD displays "Alcohol Detected".

The supplied materials do not contain quantitative calibration data, measured response time, alcohol-concentration thresholds, accuracy measurements or experimental result tables. Accordingly, no numerical performance claims are made in this documentation.

9. Engineering Considerations & Future Improvements

- Calibrate the sensing stage and establish a validated detection threshold.
- Use an ADC and analog processing if quantitative alcohol concentration measurement is required.
- Add watchdog and fault-detection mechanisms.
- Implement a controlled restart/authorization sequence after an alcohol-detected event.
- Use appropriately rated switching and safety hardware for an actual industrial machine.
- Add event logging or communication for monitoring and diagnostics.

10. Conclusion

This project demonstrates an end-to-end embedded control application using an 8051-family microcontroller. An MQ-3 sensor provides the input, the AT89C51 executes the decision logic, the L293D interfaces the controller with the motor, and the LCD and buzzer communicate the system state. The project provides practical experience in microcontroller programming, GPIO interfacing, LCD control, sensor integration and actuator control.

Appendix – Key Control Logic

```
alert = sensorIN;

if(alert != 1)
{
    motorRun();
    buzz = 0;
    cmd(0x80);
    String("Alcohol Free");
}
else
{
    motorStop();
    buzz = 1;
    cmd(0x80);
    String("Alcohol Detected");
}
```